Matrix Theory: Rank and Nullity

This study note provides a clear, practical guide to understanding the **Rank** and **Nullity** of a matrix.

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1. The Minor of a Matrix

Before understanding rank, you must define the **minor**.

- **Definition:** Take any matrix A (square or rectangular). If you delete certain rows and columns to leave behind a square sub-matrix of size $p \times p$, the determinant of that sub-matrix is called a **minor of order p**.
- **Logic:** A 3×3 matrix contains multiple 2×2 minors and one 3×3 minor (the determinant of the matrix itself).

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2. Rank of a Matrix

Represented by $\rho(A)$. The rank is the "useful" size of a matrix. It represents the number of linearly independent rows or columns.

Formal Properties

A number r is the **rank** of matrix A if it obeys these two conditions:

1. **Existence:** There is at least one minor of order r which is **not zero** (does not vanish).
2. **Vanishing:** Every minor of order higher than r is **equal to zero**.

Key Rules

Matrix Type	**Rank Property**
Zero Matrix	Always 0
Unit Matrix (I_n)	Always n
Non-Singular (det $A \neq 0$)	Rank equals the order n

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3. Nullity of a Matrix

Represented by $N(A)$. Nullity represents the "redundant" or "wasted" dimensions in a system of equations.

- **Definition:** For a square matrix of order n, the value $n - r$ is called the **nullity**.
- **Formula:**

$$N(A) = n - \rho(A)$$

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4. Practical Example: Finding Rank

While the "Minor Method" works for small matrices, the **Echelon Form** (using Row Transformations) is the professional engineering standard.

The Problem

Find the rank and nullity of the matrix A:

$$A = \begin{bmatrix} 3 & 2 & 5 \\ -1 & 0 & 6 \\ 7 & 8 & 3 \end{bmatrix}$$

Step-by-Step Transformation

To find the rank, we reduce the matrix to a form where all elements below the diagonal are zero.

Step 1: Get a '1' in the top left (a_{11}) for easier calculation.
Swap R_1 and R_2, then multiply the new R_1 by -1:

$$\begin{bmatrix} 3 & 2 & 5 \\ -1 & 0 & 6 \\ 7 & 8 & 3 \end{bmatrix} \to \begin{bmatrix} -1 & 0 & 6 \\ 3 & 2 & 5 \\ 7 & 8 & 3 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & -6 \\ 3 & 2 & 5 \\ 7 & 8 & 3 \end{bmatrix}$$

Step 2: Create zeros in the first column.
Apply $R_2 \to R_2 - 3R_1$ and $R_3 \to R_3 - 7R_1$:

$$\begin{bmatrix} 1 & 0 & -6 \\ 0 & 2 & 23 \\ 0 & 8 & 45 \end{bmatrix}$$

Step 3: Create a zero in the third row, second column.
Apply $R_3 \to R_3 - 4R_2$:

$$\begin{bmatrix} 1 & 0 & -6 \\ 0 & 2 & 23 \\ 0 & 0 & -47 \end{bmatrix}$$

Final Analysis

1. **Count Non-Zero Rows:** There are **3** non-zero rows.
2. **Rank:** $\rho(A) = 3$.
3. **Nullity:** Since the order $n = 3$, then $N(A) = 3 - 3 = 0$.

> **Summary:** Because the rank is equal to the order ($r = n$), this matrix is **non-singular**, meaning its determinant is not zero and it is fully invertible.